Team members:

Foundations

Last week we studied **simple linear regression**:

$$\hat{y}_i = \hat{b}_0 + \hat{b}_1 x_{1i}$$

Question 1

What does the estimated slope $\hat{b_1}$ represent?

Question 2

What is a residual?

We now extend to **multiple linear regression** with p predictors:

$$\hat{y}_i = \hat{b}_0 + \hat{b}_1 x_{1i} + \hat{b}_2 x_{2i} + ... + \hat{b}_p x_{2p}$$

Question 3

Suppose we are building a linear model to predict **penguin bill depth (mm)** using: x_1: flipper length (mm) and x_2: body mass (g). Why might using more than one predictor improve predictions?

Multiple predictors

Suppose we fit this model A:

$$\widehat{\text{Bill Depth}}_i = 35.5 - 0.1 \times \text{Flipper Length}_i + 0.0004 \times \text{Body Mass}_i$$

Question 4

Suppose we have flipper length and body mass for three penguins with unknown bill depth. Fill in the predicted bill depth using the estimated model above.

Penguin	Flipper Length (mm)	Body Mass (g)	Predicted Bill Depth (mm)
R	200	4000	
S	201	4000	
Q	200	4001	

Question 5

Does penguin R or S have a higher predicted bill depth? By how much was it larger?

Question 6

Does penguin R or Q have a higher predicted bill depth? By how much was it larger?

Question 7

When comparing penguins R and S, what stayed the same? _____

When comparing penguins R and Q, what stayed the same? _____

Why is this helpful when trying to understand the effect of one variable?

Question 8

When we interpret coefficients in multiple regression, we isolate the effect of one variable at a time by holding the others constant.

Complete the sentences to interpret the regression estimates:

For penguins of similar body mass, a 1 millimeter increase in flipper length is associated with a _____ mm change in predicted bill depth.

For penguins of similar flipper length, a _____ increase in body mass is associated with a _____ mm change in _____.

Categorical predictors

Now suppose we extend our model by including species as a predictor. Model B:

$$\begin{aligned}\widehat{\text{Bill Depth}}_i = &7.7 + 0.3 \times \text{Flipper Length}_i + 0.001 \times \text{Body Mass}_i \\ &- 0.2 \times I(\text{Species}_i = \text{Chinstrap}) - 5.9 \times I(\text{Species}_i = \text{Gentoo})\end{aligned}$$

Question 9

Suppose we have three penguins with a flipper length of 200 mm and a body mass of 4000 grams. Fill in the table below for Model B (one row has already been filled in for you).

Penguin	Species	$I(\text{Species}_i = \text{Chinstrap})$	$I(\text{Species}_i = \text{Gentoo})$	Predicted Bill Depth
R	Chinstrap	1	0	71.5 mm
U	Gentoo			
V	Adelie			

Question 10

Which species has the highest predicted bill depth, for penguins with similar flipper length and body mass? Which has the lowest predicted bill depth? Explain how you can tell from the model.

Evaluating models

Consider the three following models:

Model	Predictors	R^2 (train)	Adjusted R^2 (train)	RMSE (test)
Z	Flipper length	0.35	0.35	1.66
A	Flipper + Body mass	0.36	0.35	1.64
B	Flipper + Body mass + Species	0.82	0.82	0.92

Question 11

Which model described the most variation in bill depth? Which metric did you use to identify this?

Question 12

Complete the sentence: Approximately _____% of the observed variation in bill depth is explained by Model _____.

Question 13

Which model did the best predicting bill depth on new data? Which metric did you use to identify this?

Question 14

Complete the sentence: When using Model _____ on new data, predictions for bill depth were off by approximately _____ mm, on average.